Adaptive State–Reasoning Co-Design under Matched Total Compute

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Abstract

Test-time scaling is usually framed as deciding how much more reasoning to perform. We study a different decision: where computation should be spent between state construction and downstream reasoning. An initial comparison is retained as historical because its action capabilities differed. A prospectively frozen equal-action successor isolates signal complementarity: across 32 independent simulated family blocks, the two-signal policy improves exact allocation accuracy over the stronger one-signal policy by 0.253906 (stratified family-block 95% interval 0.251221–0.256653).

We then freeze one q/c/B allocation rule and apply it unchanged to SAT propagation, grid path planning and 0/1 knapsack. It matches the hindsight resource-location oracle in all nine protected cases, while reason-only and always-materialize restrictions incur positive regret in every domain. A preregistered stress study supplies the critical negative: the q-greedy rule's FLAT result replicates, but its price and distribution-shift axes are both **BROKEN**. A separately preregistered price-aware successor reaches zero regret in all 195 frozen cells when given exact published per-structure charge certificates; independent implementations agree. Formal falsifiers support the necessity/sufficiency boundary for exact certificate fields in the registered finite environment. These are bounded internal exact-domain results, not ScienceAgentBench, naturalistic-agent, forward-time-certificate, or external validation.

1 Introduction

Test-time computation has become an explicit design variable in modern reasoning systems. Systems allocate more tokens, samples, search nodes, verifier calls or iterative refinement to difficult instances. Recent work develops bandit, constrained-policy and learned adaptive allocation strategies, reinforcing a general lesson: uniform inference budgets waste computation when item difficulty is heterogeneous.

But difficulty itself has more than one source. Some tasks are difficult because the relevant structure is poorly exposed in the current representation. Others are difficult because substantial search or reasoning remains even after the right structure is visible. If a system spends all marginal budget on reasoning in an access-limited task, it reasons harder over the wrong state. If it spends all marginal budget on state construction in a reasoning-limited task, it repeatedly reorganizes information that was already accessible.

ORION-22 asks:

> **Under one matched total budget, when should a system spend computation changing state, when should it spend computation reasoning over state, and can a prospective policy learn or exploit the difference?**

The system is modeled as

`raw/current state -> optional state construction -> downstream reasoning/search -> verified outcome`.

The paper makes four contributions.

1. **Two-axis inference formulation.** State construction and downstream reasoning are symmetric budgeted actions rather than free preprocessing plus paid reasoning.
2. **A strict comparator contract.** The joint policy must beat both adaptive-state-only and adaptive-reasoning-only policies at identical total resources, not merely a fixed baseline.
3. **A comparator-capability correction.** The protected run is reported with a later hostile adjudication showing that signal count and permitted allocations varied together.
4. **An equal-action successor.** P12B holds the four actions and budget fixed, varies only visible signals, and reports family-block uncertainty.

P12A's superiority interpretation is withheld by its comparison-validity adjudication. P12B supplies the stronger controlled contract: equal total budget, equal action capability, and only then a variation in available signals. The current bounded authority is the equal-action signal-complementarity result. varied together; a capability-matched P12B is therefore required.

The historical result is intentionally controlled and exactly reproducible, but its superiority interpretation is withheld by `P12A_COMPARISON_VALIDITY_ADJUDICATION_V1.json`. Active terminal: `P12A_SUPERIORITY_AUTHORITY_WITHHELD`. Its current value is to expose the stronger contract a successor must satisfy: equal total budget, equal action capability, and only then a variation in available signals.

2 Donor boundary and novelty

2.1 Adaptive test-time compute is prior-owned

Recent systems allocate inference compute dynamically based on predicted difficulty, value or resource constraints. Bandit formulations, constrained policy optimization, adaptive demonstration/generation strategies and "when to think" policies already own the primitive that different examples deserve different reasoning budgets. ORION-22 therefore does not claim adaptive inference allocation itself.

2.2 Dynamic state construction is also prior-owned

Retrieval, compression, context selection, query-conditioned memory and structured-state construction already adapt what a model sees. ORION-21 additionally supplies controlled evidence that construction can change accessibility. ORION-22 does not claim dynamic state selection as a new primitive.

2.3 Residual after subtraction

The live residual is the **competition between those actions under one resource boundary**:

> State construction and downstream reasoning are two places to spend test-time computation. A valid joint-allocation result must hold total resource fixed and strictly improve over policies allowed to adapt either axis alone.

Current adaptive-compute donors optimize downstream reasoning, sampling, search or generation control. They do not make costed state construction and downstream reasoning symmetric decision variables under the same matched envelope, then require superiority over both one-axis adaptive controls.

3 Formal problem and why joint allocation can be strictly valuable

For item `i`, let:

- `R_i` be current/raw state;
- `c_i` be resource spent constructing/restructuring state;
- `r_i` be downstream reasoning/search resource;
- `B_i` be the total envelope;
- `z_i` be information available before the protected outcome;
- `Y_i(c_i,r_i)` be verified success or quality.

The policy chooses `(c_i,r_i)` using `z_i`, subject to the common accounting contract. In the controlled benchmark the budget is scalar and exact: `c_i+r_i<=B`. In a real system the resource is a vector and comparison is Pareto-based unless a cost scalarization is frozen before protected outcomes.

3.1 Policy classes

- `FIXED_STATE_FIXED_COMPUTE`: fixed allocation across items.
- `ADAPTIVE_STATE_ONLY`: may change state budget but not reasoning budget.
- `ADAPTIVE_REASON_ONLY`: may change reasoning budget but not state budget.
- `JOINT_STATE_REASONING`: may choose both under the same total envelope.
- `ORACLE_JOINT`: hindsight ceiling used only diagnostically.

Define
`joint_gain(B) = Q_joint(B) - max(Q_state_only(B), Q_reason_only(B))`.

A positive ORION-22 result requires `joint_gain>0` under the frozen comparison, not merely superiority to a fixed policy.

Consider a one-unit world containing two prospectively distinguishable regimes. In the access-limited regime success requires spending the unit on state construction; in the reasoning-limited regime success requires spending it on reasoning. A state-only adaptive policy cannot solve the latter and a reasoning-only adaptive policy cannot solve the former. A joint policy that sees the regime signal can spend the same unit at the valuable locus.

This existence argument is elementary and is not the empirical contribution. Its purpose is to identify the condition ORION-22 must test: **heterogeneity in the location of marginal computation value**.

4 Protected matched-budget benchmark

4.1 Resource regimes

Each protected item has one hidden requirement:

- `EASY = (0,0)`;

- `ACCESS = (2,0);`
- `REASON = (0,2);`
- `BOTH = (1,1).`

Every policy receives total budget `B=2`. Success is exact: allocated state and reasoning resources must meet both requirements. No arm receives extra budget and unused resource is not retrospectively reassigned.

4.2 Held-out families

The protected split contains **16 held-out families** × **512 items**. Family regime proportions vary, while a uniform mixture component prevents degenerate single-regime families. Signal noise `sigma_f` ranges from `0.30` to `0.80` across families.

All adaptive arms receive the same pre-outcome signals

`s_c = c_req + Normal(0,sigma_f)s_r = r_req + Normal(0,sigma_f).`

These signals contain no protected success outcome, verifier result or post-allocation feedback.

4.3 Frozen policies

- `FIXED_11`: always `(1,1)`.
- `ADAPTIVE_STATE_ONLY`: choose `(2,0)` if `s_c>=1`, else `(0,0)`.
- `ADAPTIVE_REASON_ONLY`: choose `(0,2)` if `s_r>=1`, else `(0,0)`.
- `JOINT_FROZEN`: choose the feasible allocation in {`(0,0),(1,1),(2,0),(0,2)`} nearest to `(s_c,s_r)` under squared Euclidean distance, with frozen tie order.
- `ORACLE_JOINT`: exact hindsight requirement, diagnostic only.

No policy is tuned on protected family outcomes.

5 Results

5.1 Comparator correction and equal-action successor

The historical P12A runner printed `P12A_JOINT_ALLOCATION_SUPERIORITY_SUPPORTED`, but the later capability audit found that the joint arm could emit four allocations and the named one-axis arms only two. `P12A_SUPERIORITY_AUTHORITY_WITHHELD` is therefore the binding P12A disposition.

P12B changes the estimand rather than the threshold. Every signal policy may emit the same four allocations under budget two. The independent unit is one family RNG block (n=32); 1,024 episodes in a block are technical observations. Mean two-signal gain over the stronger one-signal arm is **0.253906**, the stratified family-block 95% bootstrap interval is [**0.251221, 0.256653**], and minimum family gain is **0.196289**. An append-only locked-environment revalidation reproduces the result under CPython 3.12.13 and NumPy 2.5.2. Its terminal is `P12B_EQUAL_ACTION_SIGNAL_COMPLEMENTARITY_SUPPORTED`.

5.2 Unchanged three-domain transfer

One frozen allocator reads pending multiplicity, declared materialization cost and the shared budget; it has no domain identifier or domain-specific parameter. Across three protected cases in each of SAT propagation, 15x15 path planning and 0/1 knapsack, all arms return independently verified exact task outputs.

The numeric columns below report maximum regret for the reason-only and always-materialize controls, followed by allocator regret over the three registered cases.

domain	reason only	materialize	allocator
SAT propagation	190	2	0/3
path planning	1,202	225	0/3
knapsack	1,102	282	0/3

The terminal `P12_TRANSFER_ALLOCATION_V1_SUPPORTED` authorizes only this nine-case, three-domain result under each domain's registered charged unit.

5.3 Preregistered robustness negative

The stress study covers five build/serve price regimes, 45 original and 135 expanded case-regime cells, four case mixes and three shared-budget mixes. All task outputs remain exact, two implementations agree, and the hidden parameterization audit is green. The V1 FLAT zero-regret result replicates. However, the price-axis verdict is **BROKEN** and the distribution-shift verdict is **BROKEN**. No price regime yields zero regret over all 36 case records and the registered B1/B2 mixtures fail. This negative was retained without retuning.

5.4 Price-aware exact-certificate successor

The separately preregistered successor optimizes the same charged objective by 0/1-knapsack dynamic programming. It reads exact published per-structure reason-serve and state-serve certificates plus prices and the nominal budget. It adds no free parameter and reaches zero priced regret in all **195** frozen case/mix cells; independent exhaustive selection agrees. Its terminal is `P12_PRICE_AWARE_SUCCESSOR_SUPPORTED`.

This does not show that exact certificates are available before an action. The successor is a conditional construction-level result, not a forward-time allocator.

5.5 Selection-information boundary

The selection-sufficiency falsifier checks 7,147,140 reduced-state cells with zero mismatches and catches all three registered wrong selectors. The certificate-necessity falsifier verifies that every registered coarsening (sign, interval, threshold and certificate-free) admits an indistinguishable pair with different unique optima, while the exact-field control admits none. The terminals are `P12_SELECTION_SUFFICIENCY_THEOREM_FALSIFIER_GREEN` and `P12_CERTIFICATE_NECESSITY_THEOREM_FALSIFIER_GREEN`. This is an exact boundary inside the declared additive finite environment.

6 Resource-accounting contract and statistical analysis

The scalar controlled budget is intentionally clean. Real systems require vector receipts including:

- compiler/retrieval/preprocessing model and operations;
- state tokens/bytes and memory traffic;
- downstream generated tokens/recurrent steps;
- search nodes and verifier calls;
- tool calls and external latency;
- cache/recovery cost;
- model identity/capacity;

- end-to-end latency and reproducible energy where available.

A joint policy cannot "win" by shortening the downstream trace while hiding expensive retrieval or compilation upstream. Likewise a reasoning-only baseline cannot receive a larger model or search cap. If resource vectors are incomparable, the result should be a quality–resource Pareto frontier rather than a post-hoc weighted score.

6.1 Statistical analysis

The protected unit is the held-out family RNG block. P12A has 16 family blocks. P12B has 32 independent family blocks, eight in each fixed noise stratum. Its primary deterministic 20,000-resample bootstrap samples families within each stratum, preserving the registered mixture. An unstratified 32-family bootstrap is reported only as sensitivity.

The analysis does not pool P12A's 8,192 or P12B's 32,768 technical episodes as independent domains. Hyperparameters are frozen before protected evaluation. The worst-family and minimum-stratum gains prevent a favorable mean from hiding a family- or stratum-level failure.

Strict reconstruction also binds the execution environment. The active V1.1 receipt records the repository `uv.lock` digest, CPython 3.12.13 and NumPy 2.5.2; an environment-field mutation is a noncompensatory failure even when every scientific value is unchanged.

7 Relation to current adaptive-compute literature and limitations

Strategic test-time-compute allocation, constrained policies and "when to think" methods already own adaptive reasoning amount. Retrieval, compression and state-construction systems already own dynamic context. ORION-22's residual is joint **resource-locus allocation** under one receipt, with an equal-action capability across signal ablations.

The bounded evidence does not support a universal allocation law. The transfer set contains nine constructed exact-domain cases, and its heterogeneous charged units are not converted into a cross-domain scalar. The original q-greedy allocator is explicitly not robust to the registered price or distribution shifts.

The price-aware successor assumes exact realized additive charge certificates. Their prospective availability, acquisition cost and calibration in a deployment are not established. Its zero regret therefore cannot be promoted to a forward-time systems claim.

No artifact named P12C and no bound ScienceAgentBench/public-data result exists. External transfer across at least 20 task families, three domains and two model families remains CANNOT_CHECK. Public online data or same-owner CI would not create independent custody.

8 Discussion and conclusion

ORION-22 reframes test-time scaling as a portfolio of computations. It can be useful to build state before searching over it, but the correct locus depends on the objective and on what cost information is available.

The evidence sequence is deliberately non-monotone. P12A's apparent signal-count result is withheld because action capability differed. The equal-action P12B supports signal complementarity after matching actions. One unchanged q/c/B rule then transfers across three exact internal domains, but a preregistered stress battery breaks its price and distribution-shift claims. A price-aware successor repairs those frozen cells only after reading exact charge certificates. The formal falsifiers explain why the missing information matters.

P12*ACTIVE*CLAIM*AUTHORITY*V4.json keeps these leaves separate. The strongest current conclusion is conditional: exact charge information permits optimal selection in the registered additive setting, while coarsened or absent charge information cannot guarantee it. Whether useful certificates can be obtained prospectively on public scientific-agent tasks is the next stop/go experiment, not an already earned external result.

9 References

- Zuo, B. & Zhu, Y. *Strategic Scaling of Test-Time Compute: A Bandit Learning Approach.* ICLR 2026.
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- Zhai, Z., Li, B., Xiao, B., Li, M. & Wang, X. *Adaptive Test-Time Compute Allocation for Reasoning LLMs via Constrained Policy Optimization.* arXiv:`2604.14853`, 2026.
- *Learning When to Think: Adaptive Reasoning for Test-Time Compute Allocation.* arXiv: `2608.20256`, 2026.
- ORION-21 provides the controlled state-construction basis consumed by this paper; it does not transfer scientific authority to ORION-22.